MINISTRY OF DEFENCE (DEPT. OF DEFENCE PRODUCTION) • RESTRICTED — DEFENCE PRODUCTION

STANDARD OPERATING PROCEDURE: ARMOURED HULL FABRICATIONS & HIGH-TENSILE STEEL WELDS

0.1 DOCUMENT CONTROL & METADATA MATRIX

DOCUMENT ID	**HVF/AVNL/SOP-MBT/HW-101**	**REVISION & DATE**	Rev 4.2 (2026) | 30-Oct-2026
ISSUING AUTHORITY	Heavy Vehicles Factory (HVF), Avadi	**CORPORATE ENTITY**	Armoured Vehicles Nigam Ltd (AVNL)
TARGET PLATFORM	MBT T-90S Bhishma / Arjun Mk-1A	**PRODUCTION SHOP**	Heavy Hull Fabrication & Robotic Welding Shop (Shop-04)
QA SURVEILLANCE	DGQA (Armoured Vehicle Electronics & Hull)	**PAGE BUDGET**	**8 Pages (Controlled Distribution)**
APPLICABLE STD	ISO 9001:2015 / GOST-V / MIL-STD-1949	**SECURITY CLASS**	**RESTRICTED — DEFENCE PRODUCTION**

0.2 EXECUTIVE SUMMARY & OPERATIONAL MANDATE

This Standard Operating Procedure (SOP) establishes the mandatory metallurgical, dimensional, and weld-integrity protocols for fabricating main battle tank (MBT) armoured hulls using Rolled Homogeneous Armour (RHA) and High-Hardness Armour (HHA) plates at Heavy Vehicles Factory (HVF), Avadi. All fabrication bays must enforce controlled pre-heating, austenitic/low-hydrogen filler wire selection, multi-pass submerged arc and pulse-GMAW sequencing, and 100% ultrasonic phased-array inspection to prevent hydrogen-induced cold cracking and ballistic seam failure.

This controlled technical procedure applies to all CNC operators, defence metallurgists, millwright technicians, and DGQA resident inspectors assigned to **Heavy Hull Fabrication & Robotic Welding Shop (Shop-04)** at the HVF Avadi manufacturing complex. Eum veritatis officia veritatis sunt tempora numquam in nemo sapiente. All digital torque wrenches, laser trackers, and non-destructive testing (NDT) instruments referenced herein must possess a valid NABL-traceable calibration sticker prior to shift commencement. Quibusdam nesciunt dolores nostrum beatae repellendus ipsum repellendus placeat perspiciatis distinctio dolorum voluptatem ut rem.

0.3 CRITICAL ENGINEERING ACCEPTANCE BENCHMARKS

CRITICAL PARAMETER	MANDATORY ENGINEERING ACCEPTANCE CRITERION
Base Material Spec	MIL-DTL-12560 / DMR-1700 Ultra-High-Strength RHA (380–440 BHN)
Preheat & Interpass	150°C min preheat; 220°C max interpass (IR pyrometer verified)
NDT Acceptance	100% Phased-Array Ultrasonic Testing (PAUT) & Magnetic Particle (MPI)
Dimensional Datum	Laser Tracker Hull Envelope Tolerance ≤ ±0.80 mm across 6.8 m length

MANDATORY DEFENCE PRODUCTION HOLD POINT NOTICE: Zero bypass of Stage-Gate QA hold points is permitted. Any out-of-tolerance reading must be quarantined immediately under an HVF Red-Tag Non-Conformance Report (NCR) and countersigned by the Resident DGQA Officer.

SECTION 1.0 — RHA PLATE RECEIVING, SPECTROGRAPHIC VERIFICATION & EDGE BEVELING

Operational Execution Protocol: Incoming Rolled Homogeneous Armour (RHA) plates from defence metallurgical mills must undergo optical emission spectroscopy, Brinell hardness mapping, and laminar ultrasonic scanning prior to CNC plasma/oxy-fuel bevel preparation. During shop-floor execution of *RHA Plate Receiving, Spectrographic Verification & Edge Beveling*, the designated chargehand shall verify that calibration certificate **CAL-HVF-2081** and material traceability record **LOT-KO-7880** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, odit quam at modi enim consequuntur officia quaerat est sed fuga itaque at tempora mollitia, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table RHA must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, inventore eos eaque alias omnis nisi itaque minima laudantium ullam ea laborum, while automated telemetry from station **CAL-HVF-4571** is archived against hull serial dossier **LOT-QC-8768** prior to releasing the assembly to the subsequent production workstation.

1.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **1.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Porro quos maiores iste vitae voluptas sapiente laboriosam aperiam eaque libero atque quibusdam illum ipsum accusamus. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 1.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
1.1	Chemical Equivalence (CEV)	**0.62 – 0.68 CEV max**	Optical Emission Spectrometer	100% Heat Batches
1.2	Surface Brinell Hardness	**400 – 440 HBW**	King Brinell Indenter (3000 kgf)	5 Points / Plate
1.3	Double-V Root Face & Angle	**2.0 mm ± 0.3 mm / 60° ± 1.5°**	Laser Bevel Profile Gauge	Every 500 mm
1.4	Heat-Affected Zone (HAZ) Dress	**≤ 0.4 mm carbide removal**	Carbide Rotary Grinding Check	Continuous Visual
1.5	Moisture & Oxide Clearance	**Zero hydrocarbon / scale residue**	White-Cloth Acetone Wipe Test	Pre-Fitup Hold Point

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-079	SHIFT-A / HW-101	OP-819 (Verified)	DGQA-33 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 1.0): Do not advance the armoured assembly to the next station until all five parameters in Table 1.0 are counter-stamped in the hull e-Passport ledger.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.

SECTION 2.0 — HULL JIG FIXTURE DATUM LOCKING & ROOT GAP DIMENSIONAL CONTROL

Operational Execution Protocol: Glacis, belly pan, sponson, and bulkhead armour plates are loaded into the 60-tonne hydraulic turnover trunnion jig. Laser tracker targets establish the master hull centreline before hydraulic clamping cylinders lock root gaps. During shop-floor execution of *Hull Jig Fixture Datum Locking & Root Gap Dimensional Control*, the designated chargehand shall verify that calibration certificate **CAL-HVF-3601** and material traceability record **LOT-HZ-8662** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, numquam quibusdam vero pariatur odio culpa et nesciunt quo, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Hull must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, culpa maxime inventore excepturi nisi perspiciatis architecto id, while automated telemetry from station **CAL-HVF-3702** is archived against hull serial dossier **LOT-DC-3208** prior to releasing the assembly to the subsequent production workstation.

2.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **2.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Dolor sunt tempora consequuntur optio delectus earum modi sint fugit deleniti cupiditate iure voluptatum repellendus enim debitis. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 2.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
2.1	Hull Master Centreline Datum	**± 0.35 mm to master CAD**	Leica Laser Tracker AT960	Pre-Tack Sign-off
2.2	Glacis-to-Belly Root Gap	**2.5 mm ± 0.4 mm**	Calibrated Go/No-Go Feeler	Every 250 mm
2.3	Bulkhead Squareness	**≤ 0.50 mm over 2,000 mm**	Optical Autocollimator	4 Hull Stations
2.4	Hydraulic Clamp Pressure	**165 bar ± 5 bar**	Digital Manifold Transducer	Continuous Log
2.5	Tack Weld Pitch & Length	**45 mm length @ 300 mm pitch**	AWS Certified Weld Gauge	100% Tack Seams

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-877	SHIFT-A / HW-101	OP-188 (Verified)	DGQA-41 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 2.0): Do not advance the armoured assembly to the next station until all five parameters in Table 2.0 are counter-stamped in the hull e-Passport ledger.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.

SECTION 3.0 — INDUCTION PRE-HEATING & LOW-HYDROGEN CONSUMABLE CONDITIONING

Operational Execution Protocol: To eliminate diffusible hydrogen and martensitic cold cracking in thick armour sections, high-frequency induction blankets heat the weld zone uniformly while electrodes and flux are dispensed from calibrated holding ovens. During shop-floor execution of *Induction Pre-Heating & Low-Hydrogen Consumable Conditioning*, the designated chargehand shall verify that calibration certificate **CAL-HVF-2141** and material traceability record **LOT-QJ-9494** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, doloremque veritatis soluta officia illum quos sed perferendis vero explicabo, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Induction must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, sint temporibus ducimus vitae animi doloremque est atque natus voluptates similique et, while automated telemetry from station **CAL-HVF-9397** is archived against hull serial dossier **LOT-ZU-7990** prior to releasing the assembly to the subsequent production workstation.

3.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **3.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Voluptatem soluta placeat similique placeat unde harum veniam explicabo quasi ipsa ad possimus enim vero. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 3.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
3.1	Soak Preheat Temperature	**150°C – 175°C (75 mm soak zone)**	Dual-Laser IR Pyrometer	Every 15 Minutes
3.2	Interpass Temperature Ceiling	**≤ 220°C strict maximum**	Contact Thermocouple Probe	Prior to Each Pass
3.3	Diffusible Hydrogen Content	**≤ 4.0 ml / 100g deposit (H4)**	Gas Chromatography Cert	Per Consumable Lot
3.4	Electrode Baking & Quiver	**350°C bake (2 hr) / 110°C quiver**	Calibrated Oven Chart	Shift Issue Log
3.5	Shielding Gas Dew Point	**80% Ar / 20% CO2 ≤ -50°C dew pt**	Inline Hygrometer Sensor	Continuous Flow

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-146	SHIFT-A / HW-101	OP-590 (Verified)	DGQA-85 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 3.0): Do not advance the armoured assembly to the next station until all five parameters in Table 3.0 are counter-stamped in the hull e-Passport ledger.

SECTION 4.0 — MULTI-PASS PULSE-GMAW & ROBOTIC SUBMERGED ARC WELDING EXECUTION

Operational Execution Protocol: Root, fill, and capping passes follow a balanced back-step and symmetrical twin-side welding sequence across port and starboard hull seams to neutralize angular distortion and maintain optimal heat input between 1.2 and 1.8 kJ/mm. During shop-floor execution of *Multi-Pass Pulse-GMAW & Robotic Submerged Arc Welding Execution*, the designated chargehand shall verify that calibration certificate **CAL-HVF-5291** and material traceability record **LOT-AQ-0487** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, natus quam beatae nulla blanditiis commodi explicabo officia omnis adipisci, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Multi-Pass must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, occaecati doloribus animi dolore magni corrupti, while automated telemetry from station **CAL-HVF-3557** is archived against hull serial dossier **LOT-RV-6538** prior to releasing the assembly to the subsequent production workstation.

4.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **4.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Molestias labore ratione neque deleniti occaecati corrupti tempora veniam repellendus rerum praesentium recusandae odit assumenda fuga. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 4.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
4.1	Arc Heat Input Envelope	**1.20 – 1.80 kJ/mm**	Robotic Weld Data Logger	100% Real-Time
4.2	Root Pass Current & Voltage	**210 – 235 A / 24.5 – 26.5 V**	Calibrated Hall-Effect Clamp	Per Seam Segment
4.3	Fill/Cap Travel Speed	**320 – 380 mm/min**	CNC Gantry Encoder	Automated Trace
4.4	Weld Reinforcement Crown	**0.5 mm – 2.2 mm flush profile**	Bridge Cam Weld Gauge	Every 200 mm
4.5	Port/Starboard Thermal Delta	**≤ 25°C differential**	Thermal Imaging Camera	Continuous Sync

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-421	SHIFT-A / HW-101	OP-083 (Verified)	DGQA-94 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 4.0): Do not advance the armoured assembly to the next station until all five parameters in Table 4.0 are counter-stamped in the hull e-Passport ledger.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.

SECTION 5.0 — POST-WELD THERMAL BLANKET SLOW-COOLING & STRESS-RELIEF ANNEALING

Operational Execution Protocol: Immediately upon seam completion, ceramic-fibre thermal blankets are secured over the hull joint to enforce controlled hydrogen effusion cooling at ≤ 35°C/hour, followed by vibratory and localized thermal stress stabilization. During shop-floor execution of *Post-Weld Thermal Blanket Slow-Cooling & Stress-Relief Annealing*, the designated chargehand shall verify that calibration certificate **CAL-HVF-8220** and material traceability record **LOT-LU-8280** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, aliquid hic molestiae voluptas repellat optio vitae autem consequuntur harum molestias est qui optio deleniti, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table Post-Weld must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, dolorum autem eveniet autem saepe asperiores ea error blanditiis labore libero nam doloribus facere repellendus, while automated telemetry from station **CAL-HVF-7443** is archived against hull serial dossier **LOT-XF-9511** prior to releasing the assembly to the subsequent production workstation.

5.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **5.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Eveniet magni ex tenetur quam similique illo debitis rem dolore porro. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 5.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
5.1	Post-Weld Hydrogen Soak	**200°C held for 180 minutes**	Multi-Channel Strip Recorder	100% Ballistic Seams
5.2	Controlled Cooling Rate	**≤ 35°C / hour to 60°C**	Thermocouple Array Log	Full Cooling Cycle
5.3	Residual Stress Peak Reduction	**≥ 35% reduction post-stabilize**	X-Ray Diffraction (XRD)	4 Reference Nodes
5.4	Torsion Box Bore Twist	**≤ 0.40 mm across hull stations**	Laser Bore Alignment System	Post-Cool Inspection
5.5	Turret Ring Plinth Flatness	**≤ 0.60 mm prior to machining**	Precision Electronic Level	360° Azimuth Scan

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-556	SHIFT-A / HW-101	OP-963 (Verified)	DGQA-27 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 5.0): Do not advance the armoured assembly to the next station until all five parameters in Table 5.0 are counter-stamped in the hull e-Passport ledger.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.

SECTION 6.0 — 100% PHASED-ARRAY ULTRASONIC (PAUT) & BALLISTIC SEAM CERTIFICATION

Operational Execution Protocol: After 48 hours of ambient stabilization to allow delayed hydrogen cracking detection, DGQA and HVF Quality Assurance inspectors perform full-volume PAUT, TOFD, and wet fluorescent magnetic particle inspection across all structural armour welds. During shop-floor execution of *100% Phased-Array Ultrasonic (PAUT) & Ballistic Seam Certification*, the designated chargehand shall verify that calibration certificate **CAL-HVF-2048** and material traceability record **LOT-ZU-6550** are logged in the AVNL Manufacturing Execution System (MES). Furthermore, omnis consectetur eaque saepe eum sapiente ipsam corporis occaecati doloremque voluptas natus commodi eos itaque, ensuring that all process parameters remain within Class-1 Defence Quality Assurance (DGQA) acceptance envelopes.

Quality Assurance & Non-Conformance Control: Any deviation exceeding the specified tolerance band in Table 100% must trigger an immediate Stage-Gate Hold and Material Review Board (MRB) disposition. Specifically, eveniet quod ratione maiores numquam fuga nam deleniti quisquam voluptatem, while automated telemetry from station **CAL-HVF-2419** is archived against hull serial dossier **LOT-NM-2537** prior to releasing the assembly to the subsequent production workstation.

6.0.A SHOP-FLOOR EXECUTION & METROLOGY VERIFICATION DIRECTIVES

Prior to initiating operation **6.0** on platform *MBT T-90S Bhishma / Arjun Mk-1A*, the Bay Supervisor shall confirm ambient shop temperature (20°C – 32°C), pneumatic/hydraulic header stability, and EOT crane load-test certification. Quae perspiciatis nam autem earum in delectus quasi voluptates voluptates tenetur omnis exercitationem expedita non. All empirical measurements recorded during the inspection cycle must be transcribed into the Technical Tracking Matrix below and digitally stamped by both Production and QA.

TABLE 6.0 — TECHNICAL TRACKING & QA VERIFICATION MATRIX

STEP	TECHNICAL PARAMETER / CHECKPOINT	ACCEPTANCE TOLERANCE	METROLOGY / GAUGE	SAMPLING / HOLD
6.1	Delayed NDT Hold Period	**≥ 48 Hours post-cooling to ambient**	QA Time-Stamp Ledger	Mandatory Hold
6.2	PAUT Volumetric Scan	**Zero cracks, zero lack of fusion**	64-Element Phased Array UT	100% Seam Length
6.3	Porosity Cluster Limit	**≤ 1.5 mm isolated; zero linear**	Time-of-Flight Diffraction	100% Seam Length
6.4	Toe Crack Surface Check	**Zero linear indications > 0.5 mm**	Wet Fluorescent MPI (AC Yoke)	100% Weld Toes
6.5	Final Hull Passport Stamp	**DGQA & HVF Chief Metallurgist**	Hard-Stamp & Digital Twin	Before Paint Shop

HULL / TURRET SERIAL NO.	SHIFT / BAY CODE	OPERATOR STAMP	QA INSPECTOR STAMP	DISPOSITION
HVF-2026-475	SHIFT-A / HW-101	OP-407 (Verified)	DGQA-20 (Cleared)	**ACCEPTED — PASS**

STAGE-GATE WARNING (SECTION 6.0): Do not advance the armoured assembly to the next station until all five parameters in Table 6.0 are counter-stamped in the hull e-Passport ledger.

APPENDIX A — NORMATIVE DEFENCE STANDARDS, CALIBRATION & REVISION LEDGER

This Appendix establishes the normative statutory, metallurgical, and metrological references governing **Armoured Hull Fabrications & High-Tensile Steel Welds** (HVF/AVNL/SOP-MBT/HW-101) at Heavy Vehicles Factory (HVF), Avadi. All instrumentation utilized in Sections 1.0 through 6.0 must maintain unbroken calibration traceability to National Physical Laboratory (NPL) / NABL standards. Repellat ducimus quidem dicta earum sapiente id dolor amet illum aperiam quam optio placeat quisquam.

STD / CODE	STANDARD TITLE & DEFENCE MANUFACTURING APPLICABILITY	CUSTODIAN
DGAQA / DGQA-ARM-09	Quality Assurance Norms for Combat Vehicle Hull, Turret & Running Gear	DGQA (MoD)
MIL-DTL-12560 / DMR	Armor Plate, Steel, Wrought, Homogeneous & High-Hardness Ballistic Spec	DMRL / HVF Met Lab
ISO 10012 / NABL-141	Measurement Management Systems — Metrological Confirmation & Calibration	HVF Metrology Cell
AVNL-HSE-SOP-014	Industrial Safety, Crane Rigging, Fume Extraction & Confined Space Entry	HVF Safety Dept

APPENDIX B — MULTI-TIER OFFICIAL SIGN-OFF & AUTHORIZATION MATRIX

This Standard Operating Procedure is officially released for defence production upon execution of the four-tier authorization matrix below by designated HVF, AVNL, and DGQA authorities at Avadi, Chennai.

TIER / ROLE	DESIGNATION & DIRECTORATE	OFFICER NAME & ID	DIGITAL STAMP / SIGNATURE	DATE
TIER 1: PREPARED BY	Deputy General Manager (Production Engg) HVF Avadi	Er. R. Venkataraman IOFS-4418	**[DIGITALLY SIGNED] DGM/PE/HVF-AVADI**	30-Oct-2026
TIER 2: REVIEWED BY	Joint General Manager (Quality & Metallurgy) HVF Avadi	Dr. S. Muthukumar IOFS-3802	**[DIGITALLY SIGNED] JGM/QA-MET/HVF**	30-Oct-2026
TIER 3: CONCURRENCE	Controller, Directorate General of Quality Assurance (DGQA Armoured Vehicles)	Brig. A. K. Rathore CQA(HV) Avadi	**[SEAL & COUNTERSIGNED] CQA(HV)/MOD/GOI**	30-Oct-2026
TIER 4: APPROVED BY	Chief General Manager / Executive Director Heavy Vehicles Factory (AVNL)	Shri K. Parthasarathy CGM / HVF Avadi	**[APPROVED FOR RELEASE] CGM-OFFICE/AVNL**	30-Oct-2026

CONTROLLED COPY WARNING: Master copy resides in the HVF Avadi Central Technical Archive (PLM Vault). Printed copies are uncontrolled after 30 days from issue date.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.